

LUCIDMSK CME PROGRAM

MODULE 06 — POST-TEST

Wrist & Hand MRI: Ligaments, Tendons & Classification Systems

Wrist/Hand · 10 Questions · 1.0 AMA PRA Cat 1 Credit(s)[™]

Read each question carefully and select the single best answer. A score of $\geq 70\%$ (7/10 correct) is required to claim CME credit. The complete answer key with detailed explanations begins after Question 10.

POST-TEST QUESTIONS

1 Scapholunate gap measures 4.5 mm on PA wrist radiograph. This is called:

- A Normal variant
- B Terry Thomas sign — scapholunate dissociation
- C Gilula line disruption only
- D Physiologic laxity

2 Thumb UCL torn stump is displaced SUPERFICIAL to the adductor aponeurosis. This is:

- A Grade II partial tear
- B Grade III complete (non-Stener)
- C Stener lesion — surgical repair required
- D Grade I sprain

3 TFCC Palmer Type 1B tear involves:

- A Central disc perforation
- B Peripheral detachment at the ulnar styloid/fovea
- C Radial avulsion at sigmoid notch
- D Distal radioulnar ligament with DRUJ instability

4 Lichtman Stage IIIA Kienbock disease. Defining feature vs. IIIB:

- A Lunate signal change only
- B Lunate collapse without fixed scaphoid rotation
- C Lunate collapse WITH fixed scaphoid rotation (DISI)
- D Radiocarpal arthrosis

5 MRI shows A2 pulley thickening with bowstringing >3 mm from bone on axial MRI. Severity:

- A Grade 1 — sprain
- B Grade 2 — partial; <3 mm bowstringing
- C Grade 3 — complete; >3 mm bowstringing
- D Grade 4 — multiple pulleys

6 Best imaging sequence/plane for ECU subluxation in a tennis player:

- A Coronal PD fat-sat
- B Axial PD fat-sat through the ulnar styloid
- C Sagittal STIR
- D 3D GRE

7 Geissler Grade III SL ligament tear at arthroscopy:

- A Normal
- B Partial; no probe passage
- C Partial/complete; probe passes between SL and lunate
- D Complete; 2.7 mm scope passes through gap

8 Eaton-Littler Stage II CMC osteoarthritis. Findings:

- A Normal ≤ 1 mm joint space
- B Joint narrowing + osteophytes; NO subluxation
- C Narrowing + osteophytes + subluxation <1/3 joint
- D Subluxation >1/3 + pantrapezial arthrosis

9 Mayfield Stage II perilunate instability involves disruption of:

- A SL ligament only

- | | |
|----------|---------------------------------|
| B | SL + lunocapitate (midcarpal) |
| C | SL + LT ligament |
| D | Complete perilunate dislocation |

10 Best MRI sequence/plane to identify a Stener lesion:

- | | |
|----------|---------------------------------------|
| A | Sagittal T1 fat-sat |
| B | Axial PD through metacarpal head |
| C | Coronal T1 or PD fat-sat at MCP level |
| D | Sagittal STIR |
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ANSWER KEY & EXPLANATIONS

Q	1	2	3	4	5	6	7	8	9	10
ANS	B	C	B	B	C	B	C	C	B	C

Q1 Scapholunate gap measures 4.5 mm on PA wrist radiograph. This is called: **Answer: B**

A Normal variant

✓ **Terry Thomas sign — scapholunate dissociation**

C Gilula line disruption only

D Physiologic laxity

EXPLANATION Terry Thomas sign = SL gap >3 mm on PA wrist, indicating scapholunate ligament disruption. Named after a gap-toothed British comedian. This is the key radiographic sign of SL dissociation.

Q2 Thumb UCL torn stump is displaced SUPERFICIAL to the adductor aponeurosis. This is: **Answer: C**

A Grade II partial tear

B Grade III complete (non-Stener)

✓ **Stener lesion — surgical repair required**

D Grade I sprain

EXPLANATION Stener lesion = torn proximal UCL stump displaced superficial to (folded over) the adductor aponeurosis. This prevents healing — surgical repair is required. Non-Stener Grade III may heal conservatively.

Q3 TFCC Palmer Type 1B tear involves: **Answer: B**

A Central disc perforation

✓ **Peripheral detachment at the ulnar styloid/fovea**

C Radial avulsion at sigmoid notch

D Distal radioulnar ligament with DRUJ instability

EXPLANATION Palmer Type 1B = peripheral ulnar-sided detachment at the ulnar styloid/fovea — traumatic, most common repairable type. Type 1A = central disc perforation (non-repairable). Type 2 = degenerative.

Q4 Lichtman Stage IIIA Kienbock disease. Defining feature vs. IIIB: **Answer: B**

A Lunate signal change only

✓ **Lunate collapse without fixed scaphoid rotation**

C Lunate collapse WITH fixed scaphoid rotation (DISI)

D Radiocarpal arthrosis

EXPLANATION IIIA = lunate collapse WITHOUT fixed scaphoid rotation. IIIB = same but WITH fixed DISI deformity — the critical distinction for surgical planning. Stage II = early collapse. Stage IV = arthrosis.

Q5 MRI shows A2 pulley thickening with bowstringing >3 mm from bone on axial MRI. Severity:

Answer: C

- A Grade 1 — sprain
- B Grade 2 — partial; <3 mm bowstringing

✓ **Grade 3 — complete; >3 mm bowstringing**

- D Grade 4 — multiple pulleys

EXPLANATION Grade 3 = complete A2 pulley rupture with >3 mm bowstringing. Grade 2 = partial tear with <3 mm. Grade 4 = combined A2 + adjacent pulley injury. >3 mm bowstringing is the key MRI criterion.

Q6 Best imaging sequence/plane for ECU subluxation in a tennis player:

Answer: B

- A Coronal PD fat-sat

✓ **Axial PD fat-sat through the ulnar styloid**

- C Sagittal STIR
- D 3D GRE

EXPLANATION Axial PD fat-sat through the ulnar styloid in supination is key for ECU subsheath integrity and tendon position. ECU subluxation is position-dependent and may be missed in neutral.

Q7 Geissler Grade III SL ligament tear at arthroscopy:

Answer: C

- A Normal
- B Partial; no probe passage

✓ **Partial/complete; probe passes between SL and lunate**

- D Complete; 2.7 mm scope passes through gap

EXPLANATION Geissler Grade III = probe (but NOT the arthroscope) can be passed between scaphoid and lunate. Grade IV = scope (2.7 mm) passes through. Grade I–II = partial tears with no instability.

Q8 Eaton-Littler Stage II CMC osteoarthritis. Findings:

Answer: C

- A Normal ≤ 1 mm joint space
- B Joint narrowing + osteophytes; NO subluxation

✓ **Narrowing + osteophytes + subluxation <1/3 joint**

- D Subluxation >1/3 + pantrapezial arthrosis

EXPLANATION Eaton Stage II = joint space narrowing + sclerosis + osteophytes + subluxation <1/3 of the joint. Stage I = normal. Stage III = subluxation >1/3. Stage IV = pantrapezial arthrosis.

Q9**Mayfield Stage II perilunate instability involves disruption of:****Answer: B**

- A SL ligament only
- ✓ **SL + lunocapitate (midcarpal)**
- C SL + LT ligament
- D Complete perilunate dislocation

EXPLANATION Mayfield Stage II = SL + lunocapitate (midcarpal) disruption. Stage I = SL only. Stage III = SL + LT. Stage IV = complete perilunate/lunate dislocation. Injury propagates radial to ulnar.

Q10**Best MRI sequence/plane to identify a Stener lesion:****Answer: C**

- A Sagittal T1 fat-sat
- B Axial PD through metacarpal head
- ✓ **Coronal T1 or PD fat-sat at MCP level**
- D Sagittal STIR

EXPLANATION Coronal T1 or PD fat-sat at the 1st MCP level is optimal. The Stener lesion appears as the UCL stump folded back superficial to the adductor aponeurosis — the classic "yo-yo on a string" appearance.